

2110446 DATA SCIENCE AND DATA ENGINEERING

Homework 9: Spatial Analysis

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1 Source Code

The modified `streamlit_airbnb.py` can be accessed via this link https://drive.google.com/file/d/1VbKk_hfcBod0N9phr95emXl3ewMm0Haf/view?usp=sharing or an attached file in the MyCourseVille submission.

Note that the Mapbox is now requires the API key to be able to access the map visualization. Make sure to create a `/.streamlit/secrets.toml` file to make the page able to show the map beneath the scatter or heatmap plot.

API Key Configuration

```
1 MAP_BOX_API_KEY = "[INSERT YOUR API TOKEN HERE]"
```

You can create your own API token by create a free account at the website <https://www.mapbox.com>

To utilize the API key, please insert the `api_keys` parameter on the PyDeck function. If you don't configure the API keys and passing the API key to the PyDeck parameter, the map layer won't appear on the screen.

Utilize the API Keys

```
1 st.pydeck_chart(  
2     pdk.Deck(  
3         api_keys={"mapbox": st.secrets.get("MAP_BOX_API_KEY")},  
4         layers=[heatmap_layer],  
5         initial_view_state=pdk.ViewState(  
6             latitude=filtered_data["latitude"].mean(),  
7             longitude=filtered_data["longitude"].mean(),  
8             zoom=11,  
9             pitch=0,  
10        ),  
11        map_style=MAP_STYLES[map_style],  
12    ),  
13    height=600,  
14 )
```

2 Map Visualization

In the Streamlit page, there are total 4 map visualizations which are.

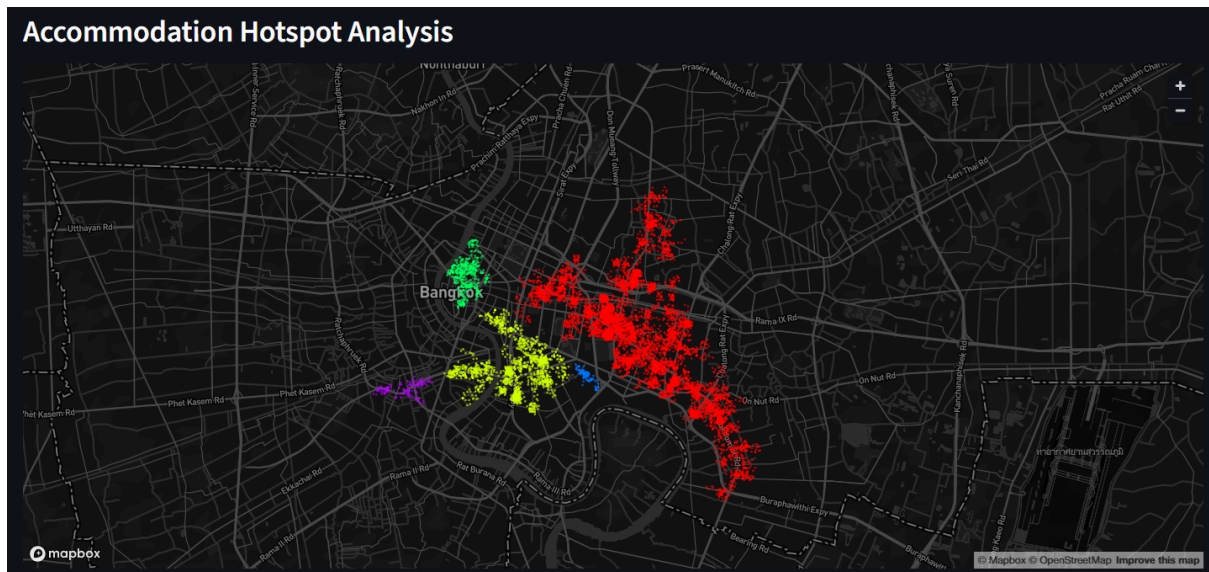


Figure 1: Accommodation Hotspot Clusters

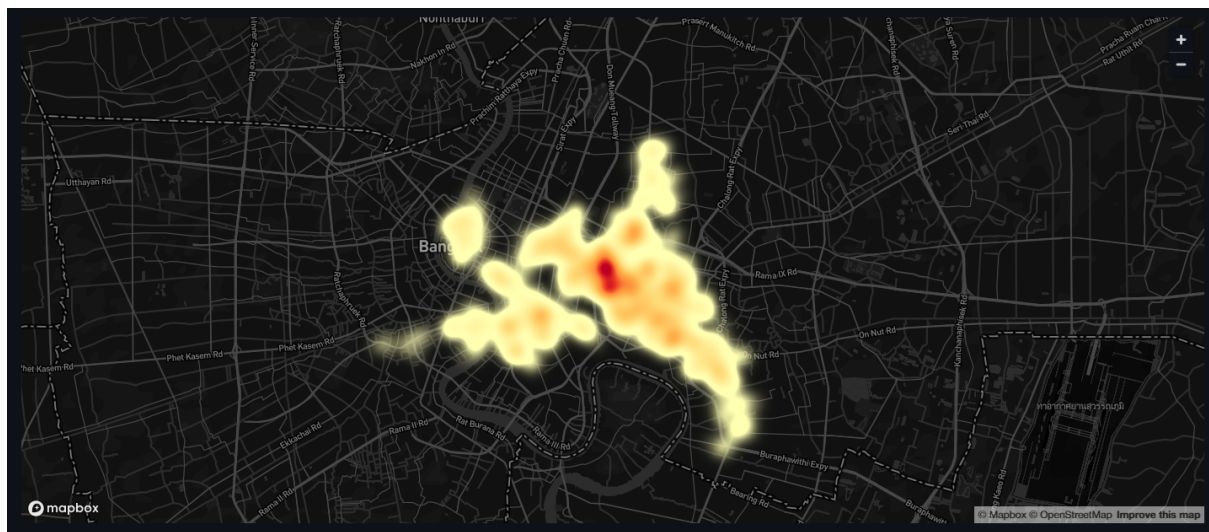


Figure 2: Accommodation Hotspot Heatmap

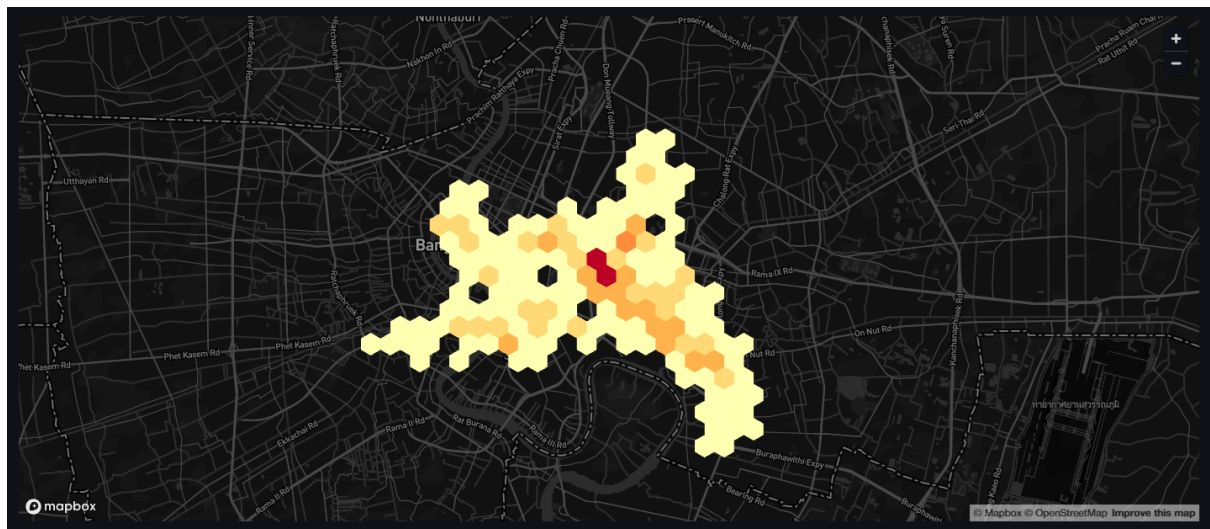


Figure 3: Accommodation Hotspot Hexagon Heatmap

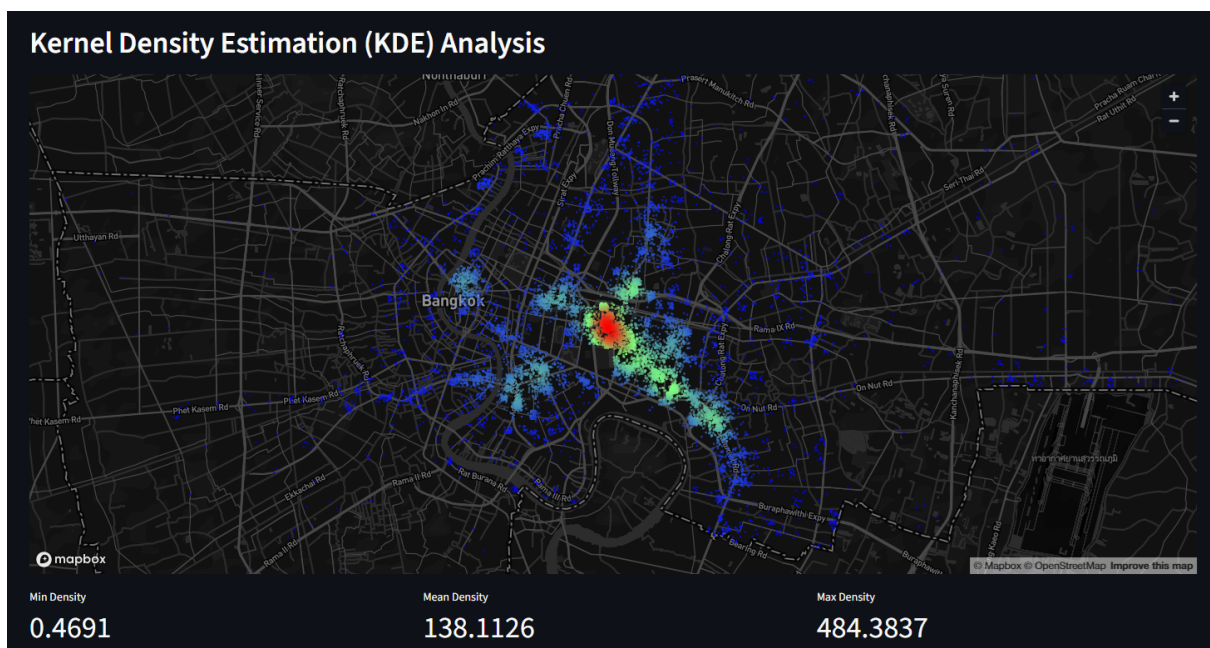


Figure 4: Kernel Density Estimation (KDE) Analysis

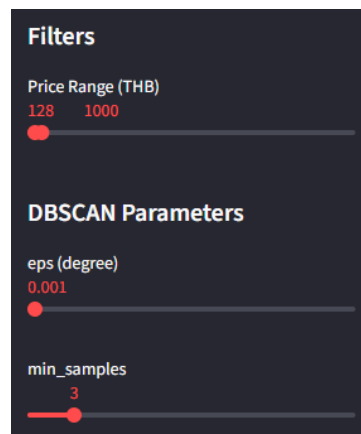
3 Answers

Consider only the accommodation which price is not exceed 1,000 baht. Answer the following questions

3.1 Problem I

According to the DBSCAN clustering by setting parameters `eps` to 0.001 (approximately 111 meters) and `min_samples` to 3. Which district has the biggest cluster?

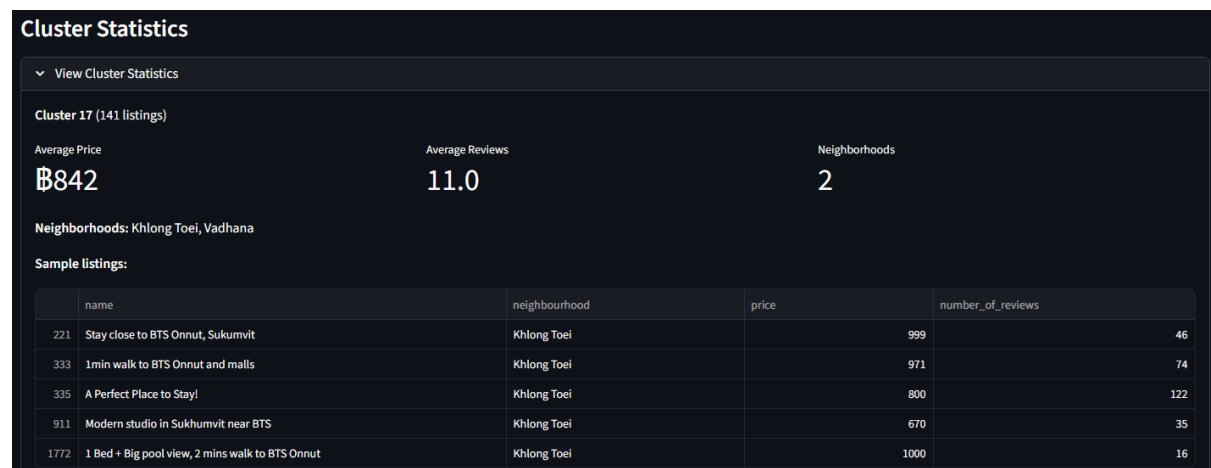
First, set the parameters to this at the sidebar of the Streamlit page.



The screenshot shows a sidebar with two sections: 'Filters' and 'DBSCAN Parameters'. In the 'Filters' section, the 'Price Range (THB)' is set with a slider from 128 to 1000. In the 'DBSCAN Parameters' section, 'eps (degree)' is set to 0.001 and 'min_samples' is set to 3.

Figure 5: Parameters Configuration

Then, go to the cluster statistics section and look at the first one.



The screenshot shows the 'Cluster Statistics' section. It displays 'Cluster 17 (141 listings)' with an 'Average Price' of ฿842, 'Average Reviews' of 11.0, and 'Neighborhoods' of 2. Below this, it lists 'Neighborhoods: Khlong Toei, Vadhana' and 'Sample listings:'. A table follows with columns: name, neighbourhood, price, and number_of_reviews.

	name	neighbourhood	price	number_of_reviews
221	Stay close to BTS Onnut, Sukumvit	Khlong Toei	999	46
333	1min walk to BTS Onnut and malls	Khlong Toei	971	74
335	A Perfect Place to Stay!	Khlong Toei	800	122
911	Modern studio in Sukhumvit near BTS	Khlong Toei	670	35
1772	1 Bed + Big pool view, 2 mins walk to BTS Onnut	Khlong Toei	1000	16

Figure 6: Cluster Information

Ans. Khlong Toei district and Vadhana district has the biggest cluster.

3.2 Problem II

According to the KDE analysis by setting parameters **bandwidth** to 0.005 (approximately 555 meters) as default. What is the accommodation name at the highest density point?

First, set the parameters to this at the sidebar of the Streamlit page.

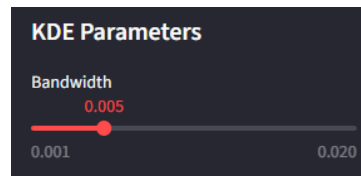


Figure 7: Parameters Configuration

Then, go to the KDE top 10 density listings right below the map section and look at the first one.

Min Density		Mean Density		Max Density	
1.4179		84.9882		276.9042	
	name	neighbourhood	price	density	number_of_reviews
0	Sukhumvit Suite: Onnut BTS, Malls & Street Food	Khlong Toei	930	276.9042	51
1	room w/2 beds private bathroom	Khlong Toei	450	276.5245	2
2	1 Bed + Big pool view, 2 mins walk to BTS Onnut	Khlong Toei	1000	276.5215	16
3	Metro Oasis On Nut/BTS/WIFI/Pool/Gym	Khlong Toei	800	276.0846	15
4	271Near BTS#Comfortable condo#Free Wi-Fi#Pool#GYM	Khlong Toei	950	276.0232	11
5	On nut Rhythm Sukhumvit 50, 1BR Fully Furnished	Khlong Toei	890	275.9797	11
6	Modern studio in Sukhumvit near BTS	Khlong Toei	670	275.4571	35
7	素坤逸50巷bts onnut酒店式公寓【开间双床房】	Khlong Toei	950	275.376	9
8	4 mins walk to BTS Onnut Sukhumvit 52	Khlong Toei	650	274.9306	1
9	素坤逸50巷 酒店式公寓双床房【开间】 bts onnut	Khlong Toei	950	274.2759	24

Figure 8: Top 10 Density Listings

Ans. Sukhumvit Suite: Onnut BTS, Malls & Street Food.